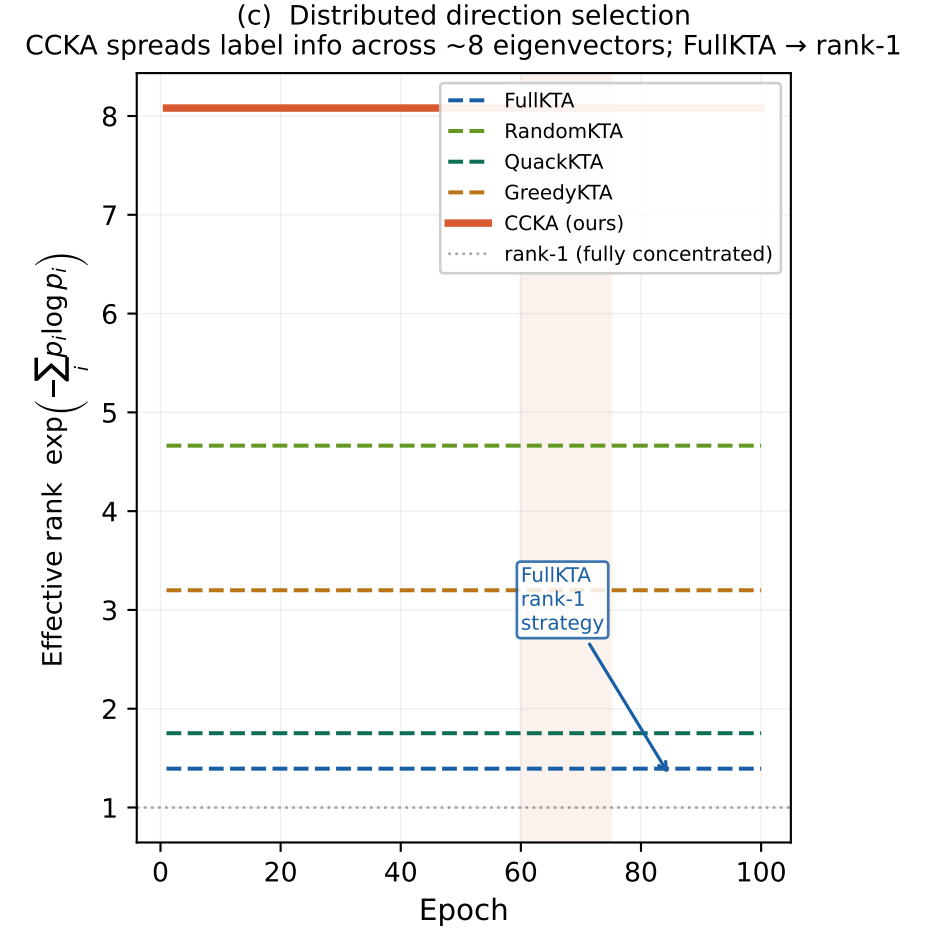
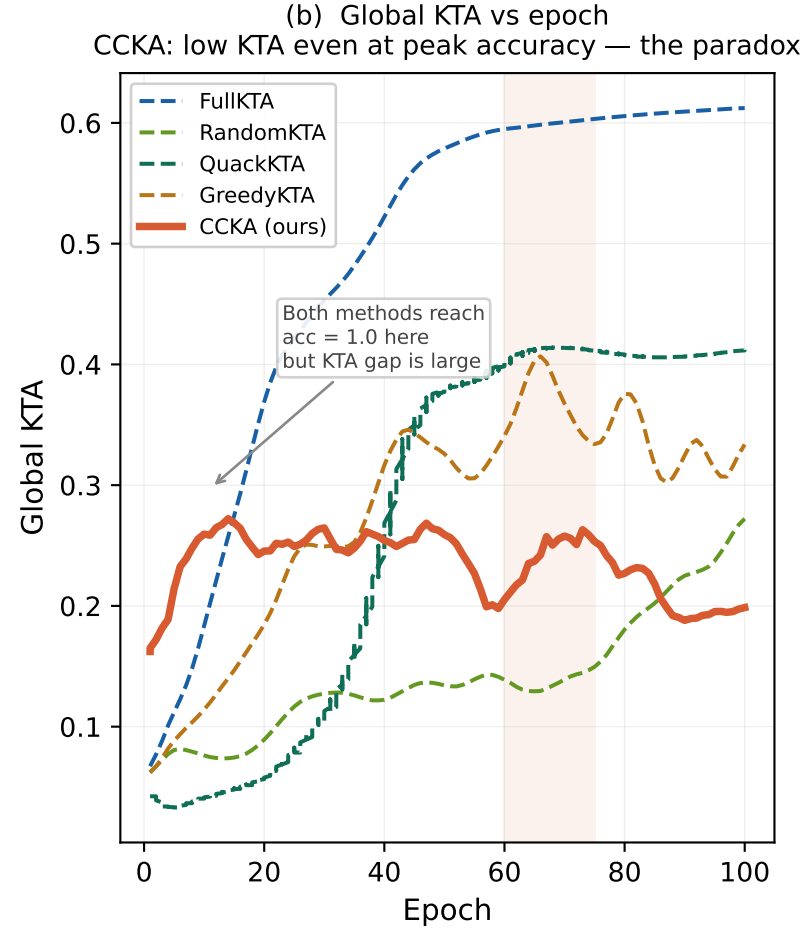
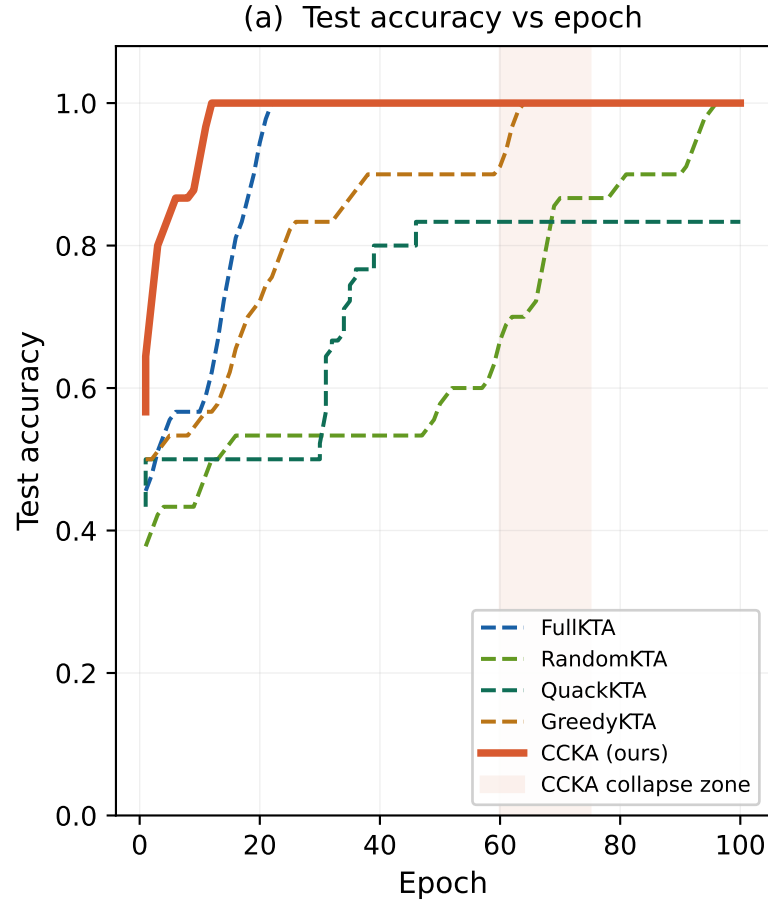


Behavioural analysis of CCKA — CHECKERBOARD

Fast convergence via distributed multi-direction kernel alignment



(a) CCKA reaches peak accuracy faster than all baselines at lower computational cost, but without FullKTA's stability. (b) CCKA maintains low global KTA even at peak accuracy — the standard metric fails to capture its success. (c) The resolution: FullKTA concentrates all discriminative information into a single kernel eigenvector (rank $\rightarrow 1$, $p_1 \approx 1$), while CCKA distributes it across ~8 directions simultaneously (high effective rank). Both strategies achieve perfect classification but via opposite spectral geometries.